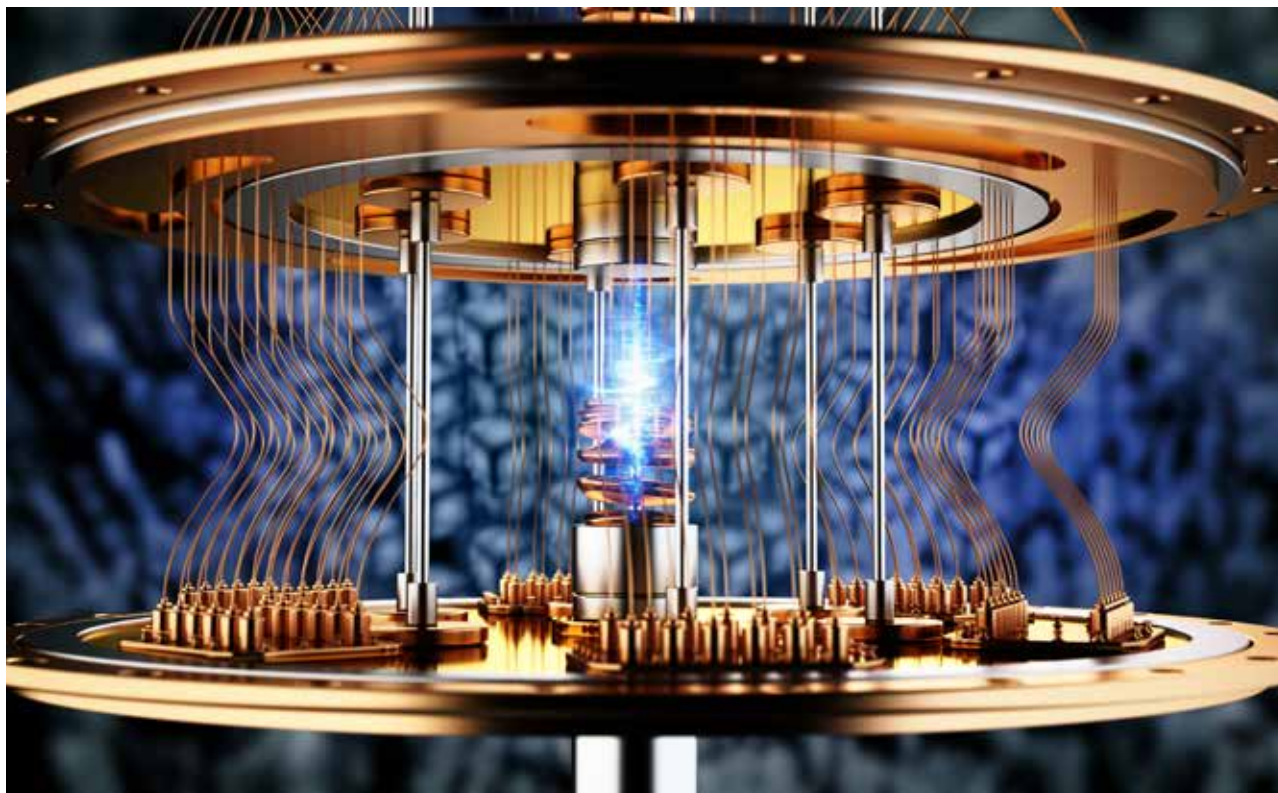


Quantum 2.0: The Next Big Tech Revolution



Quantum tech is no longer science fiction. From computing to cryptography, here's how India is gearing up for a quantum-powered future.

The United Nations has designated 2025 as the International Year of Quantum Science and Technology (IYQ 2025) to mark 100 years of quantum mechanics and spotlight its transformative impact. A series of global events will showcase how quantum technologies are revolutionising secure communication, advanced computing, ultra-precise sensing, and metrology. As the world steps into the era of the Second Quantum Revolution—or Quantum 2.0—breakthroughs in

quantum computing, cryptography, and communication are poised to reshape industries and tackle critical global challenges.

Amid shifting geopolitical dynamics, digital sovereignty has become a strategic priority. Nations leading in quantum technologies are expected to gain a decisive edge. Global powers are investing heavily to stay ahead in the innovation race. According to Qureca, worldwide investments in quantum science and technology have surpassed \$40 billion, with

China alone contributing around \$15 billion. McKinsey & Company's 2023 Quantum Technology Monitor Report projects the market will reach \$106 billion by 2040. Quantum science is set to unlock new frontiers—redefining innovation and placing itself at the core of the next technological revolution.

First quantum revolution

The first quantum revolution, which began in the early 20th century, laid the foundation for modern physics by introducing the principles of